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SUDHEER K

Senior Software Engineer

SUMMARY

10 years of experience in backend software development and systems architecture, specializing in full lifecycle projects including requirement gathering, coding, testing, and deployment. Proven ability in project management, cross-functional team leadership, and delivering complex, scalable systems across web, mobile, and on-premise platforms. Led architecture of an enterprise on-prem VM platform with reproducible builds and fail-safe OTA updates

TECHNICAL SKILLS

Server Side Scripting:

FastAPI, Python-Django, Flask, Django Rest Framework

Mobile App Development:

Flutter for Android/iOS, Android [Java].

Front End:

ReactJs, AngularJs, HTML, CSS, JavaScript, JQuery

Cloud, DevOps & Tools :

Packer, AWS, GCP, Git, Docker, Jenkins, Linux, Selenium, Scrapy, Capistrano, Firebase, Elasticsearch, RabbitMQ, Celery, PostgreSQL, Socket

Payment Integrations:

Stripe, Mollie, Razor pay

PROFESSIONAL EXPERIENCE

Cambium Networks, Kochi — *Sr. Engineer*

AUGUST 2022 - PRESENT

- Contributed to the development of LINKPlanner (web & desktop), implementing standalone features such as ANP, GIS Layers, and Custom Antennas to enhance large-scale network planning workflows.
- Architected the On-Prem LINKPlanner solution by packaging the web application into a Debian-based VM for enterprise deployments, allowing security-sensitive clients to run fully isolated workflows.
- Designed and built a reproducible VM image creation pipeline using Packer.
- Implemented an automated CI/CD pipeline using Jenkins and AWS EC2 for VM builds and artifact distribution.
- Developed a fail-safe OTA update mechanism using RAUC, enabling seamless remote upgrades.
- Designed asynchronous processing pipelines using Celery to improve system responsiveness.

Tech Stack Environment: Python, FastAPI, React, Debian, Packer, Jenkins, AWS, RAUC, Celery, PostgreSQL.

Nuventure Connect, Kochi

AUGUST 2021 - AUGUST 2022 — *TeamLead*

- Led a cross-functional team (backend, frontend, mobile) of 6 members delivering multiple SaaS and IoT products.
- Owned architecture decisions, project estimations, system design, and production deployments.
- Improved code quality and delivery speed through structured code reviews and task planning.
- Delivered multiple production systems with high reliability and maintainability.

Tech Stack Environment: Python, Django, Flutter, React, AWS, Docker.

FEBRUARY 2016 - AUGUST 2021 — *Senior Software Developer*

- Built and scaled 10+ applications across web and mobile platforms, handling Server-Side Development, Mobile App Development, Frontend, and Server Deployment.
- Designed REST APIs and backend systems using Django and Flask.
- Implemented background job processing using Celery and RabbitMQ.
- Integrated third-party systems including payment gateways (Stripe, Mollie) and external APIs.
- Developed and optimized a high-traffic CMS backend handling ~2-3 million requests/day using Flask on GCP with Varnish caching and Sphinx search.
- Reduced page load time from ~30-40 seconds to <3 seconds by instrumenting code with detailed timing metrics and implementing targeted performance optimizations.

Tech Stack Environment: Python, Flask, Django, Celery, RabbitMQ, PostgreSQL, GCP, Android (Java), ReactJS, AngularJS.

ACHIEVEMENTS/AWARDS

Spot Award: JANUARY 2023 & JULY 2025 Cambium Networks.

Employee of the Month: JANUARY 2021 Nuventure Connect.

Best Performer: APRIL 2022 Nuventure Connect.

KEY PROJECTS

LINKPlanner, Cambium Networks.

LINKPlanner allows you to model “what if” scenarios for fixed wireless technologies – based on geography, distance, antenna height, transmit power, and other factors – to optimize system performance before purchase. Quickly design networks for optimal deployment and cost effectiveness with ease.

- **Role:** Backend and Front End Development. Developed standalone features like ANP, GIS Layers, CNMaestro Integration, Incorporating new cambium products and other features.
- **Languages/Tools:** Python, FastAPI, React, Celery, Postgresql, Jenkins, AWS

On Prem LINKPlanner, Cambium Networks

A standalone, Debian-based Virtual Machine (OVA) distribution designed for enterprise environments to provide wireless modeling in a secure, self-contained virtualization format.

- **Role:** Lead Architect & Systems Engineer. Designed the end-to-end architecture, automated CI/CD pipelines, reproducible builds via Packer, and fail-safe updates via RAUC.
- **Languages/Tools:** Python, Debian, Packer, RAUC, Jenkins, AWS (EC2, S3), VirtualBox, Shell Scripting

Smart Garden, Nuventure Connect.

An interface for users to control the smart garden controller. The mobile app connects to the Web server via APIs to authenticate users and access resources like Programs and station configs.

- **Role:** Project Estimation, Architecture Designing, Building backend and Mobile Application, Deployment.
- **Languages/Tools:** Python, Django, Django Rest Framework, Celery, HTML, CSS, Ajax, JQuery, Flutter, OpenSprinkler API

Product Information Portal (PIP) Tool, Nuventure Connect.

A web-based interface for staff to see details of products handled daily, showing data in Perfion, organized per parent on one page.

- **Role:** Project Estimation, Architecture Designing, Building backend, Deployment.
- **Languages/Tools:** Python, Django, Celery, Zeep, XlsxWriter, HTML, CSS, Ajax, JQuery.

MyWork App, Nuventure Connect.

Brings together everything a team needs to communicate, collaborate, and coordinate work.

- **Role:** Backend development team. Implemented Subscriptions using Stripe, Email automation, Geo locations, Customisations in Django admin.
- **Languages/Tools:** Python, Django, Django Rest Framework, Stripe, Celery, HTML, CSS.

OneBoard, Nuventure Connect.

A cloud-based dashboard and rule engine for viewing/monitoring data from devices connected to any controller, enabling cross-controller and third-party API automation.

- **Role:** Project Estimation, Architecture Designing, Database Design, Building backend, Deployment.
- **Languages/Tools:** Python, Django, Django Rest Framework, Celery, HTML, CSS, Ajax, JQuery, UbiDot

PoolLux, Nuventure Connect.

A mobile application helping users register devices and request warranties or services.

- **Role:** Project Estimation, Architecture Designing, Building backend and Client Application, Deployment.
- **Languages/Tools:** Python, Django, Django Rest Framework, Celery, HTML, CSS, Ajax, JQuery, Flutter, OpenSprinkler API

SpinWheel, Nuventure Connect.

Marketing application to offer attractive prizes or discounts to website visitors via a freebie wheel.

- **Role:** Architect: Handled project design, estimation, code reviews, and deployment.
- **Languages/Tools:** Python, Django, Celery, HTML, CSS, Ajax, JQuery, RazorPay

Pool Configurator, Nuventure Connect.

Web app built in ReactJs helping customers enquire and generate an estimate of desired pools.

- **Role:** Project Estimation, Architecture Designing, Building backend and Client Application, Deployment.
- **Languages/Tools:** Python, Django, Django Rest Framework, Celery, HTML, CSS, ReactJs

NorsupOne, Nuventure Connect.

Web app developed for MegaGroup to help swimming pool installers ship chemical products directly from the supplier to end-users.

- **Role:** Developed the Web Shop and REST API for the mobile application.
- **Languages/Tools:** Python, Django, Rest API, AX-API, Mollie, OAuth2, Rollbar, HTML, CSS, Whoosh, Haystack, Celery, CKEditor, Pillow.

NorsupOne Mobile, Nuventure Connect.

A mobile app designed to control Smart Controllers and related devices (device provisioning, smart controls, sensor calibration).

- **Role:** Developing the mobile application.
- **Languages/Tools:** Flutter, Dart.

Chemical Resistance, Nuventure Connect.

A tool built in 24 hours to check if an item is suitable for a given chemical or fluid.

- **Role:** Developing the whole backend logic.
- **Languages/Tools:** Python, Django, HTML, CSS.

Aqua Innovation, Nuventure Connect.

A forum where Innovators meet Customers and Professionals.

- **Role:** Developing the whole backend logic.
- **Languages/Tools:** Python, Django, Django Machina, HTML, CSS, Whoosh, Haystack, Celery.

Mediaone, Nuventure Connect.

Malayalam news channel broadcasting in India and the Middle East.

- **Role:** Creating RESTful APIs for web/mobile services and writing front-end logic using AngularJS.
- **Languages/Tools:** Python, Django, Django Rest Framework, AngularJS, HTML, CSS, Elasticsearch, Amazon S3, RabbitMQ, Celery.

Heat Pump Selection Tool, Nuventure Connect.

Mobile app to select the best/economical heat pump for a pool of given specifications.

- **Role:** Project Estimation, Architecture Designing, Building backend and Mobile Application, Deployment.
- **Languages/Tools:** Python, Django, Django Rest Framework, Celery, HTML, CSS, Flutter.

Service Scanner, Nuventure Connect.

Handy app for MegaGroup Service center to view/update tickets and Scan/Generate QR codes for label printers.

- **Role:** Project Estimation, Architecture Designing, Coding.
- **Languages/Tools:** Flutter, Dart.

ClubMe, Nuventure Connect.

Trainer app for finding trainers nearby and getting trained.

- **Role:** Implemented features for the Android app and web page for Rich Text content post creation.
- **Languages/Tools:** Android, Python, Firebase, Django, HTML, CSS, RabbitMQ, Celery.

HOBBY/LEARNING PROJECTS

Covin Vaccinate Me

Covin is a simple application that helps users to get notified when a vaccine is available in their vaccination centres of interest.

This project has

- **Covin Bot**
A telegram bot that helps to search for vaccine availability by the district as well as Pincode and has an option to subscribe for notifications
- **Covin Webapp**
Webapp that helps to search for vaccine availability and also save a search.

Code:

- Telegram bot: <https://github.com/zudheer/cowin-bot>
- Web app: <https://github.com/zudheer/cowin-web>

Toy Linux Shell

A simple Linux shell written in python that can execute Linux terminal commands uses python system calls like 'execvp'.

Code: <https://github.com/sudheerkdty/toy-python-shell>

Translating Peter Norvig's Tiny Lisp (in Python) to Javascript

A lisp interpreter in javascript with basic interpreter functions.

Code: <https://github.com/sudheerkdty/LISP-Interpreter->

HTML Graph Canvas

A canvas to mark nodes and draw vertices and then represent graph algorithms. The intention was to familiarize me with javascript.

Code: https://github.com/sudheerkdty/graph-html_canvas

ACADEMIC PROJECTS

Semantic Information Retrieval System - Main Project

The goal is to implement an indexing structure in which the index is built on the basis of context and semantics of the document rather than on the terms.

Multiple Sequence Matching Using Boolean Algebra - Mini Project

The aim of the project was to implement sequence-matching algorithms that compute the degree of similarity or dissimilarity between gene sequences.

EDUCATION

B.Tech, Computer Science & Engineering

University of Calicut, Kerala - SGPA 6.54

AUGUST 2011 - AUGUST 2015

Govt. Engineering College, Sreekrishnapuram, Palakkad

Higher Secondary, Science,

HSE Kerala, India - 85.58%

AUGUST 2008 - MARCH 2010

Karimpuzha Higher Secondary School, Thottara, Palakkad

High School,

Central Board of Secondary Education, India - 85.8%

March 2008

St Dominics' Convent EMHSS, Sreekrishnapuram, Palakkad

LANGUAGES

English (Professional Working Proficiency)

Malayalam (Native)

Hindi (Can read and write; limited spoken proficiency)